



MOSAIC: A Spectral Framework for Integrative Phenotypic Characterization Using Population-Level Single-Cell Multi-Omics

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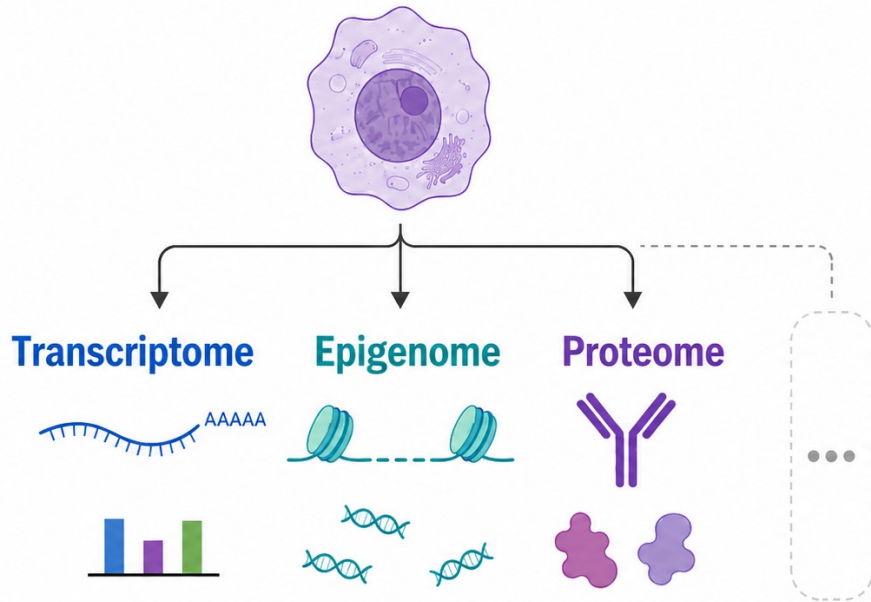
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* These authors jointly supervised this work.

Single-Cell Studies Are Expanding Along Three Critical Axes

Multi-Modality



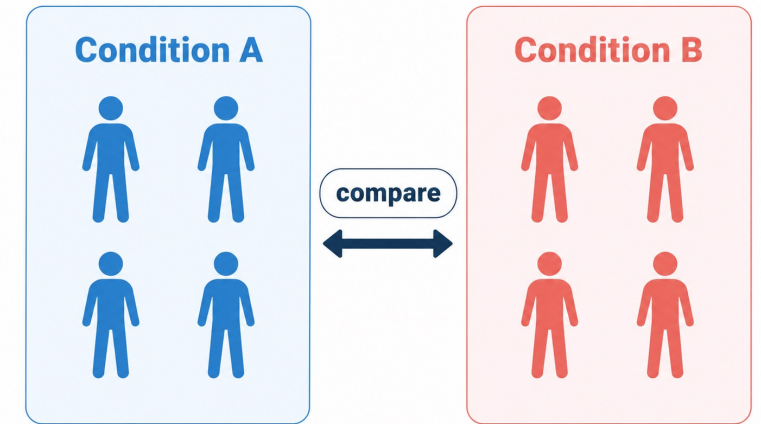
- Multiple molecular modalities profiled from the same cell, capturing rich cross-modal regulatory relationships

Multi-Individual



- Cohorts now span dozens to hundreds of individuals, each with a unique molecular profile

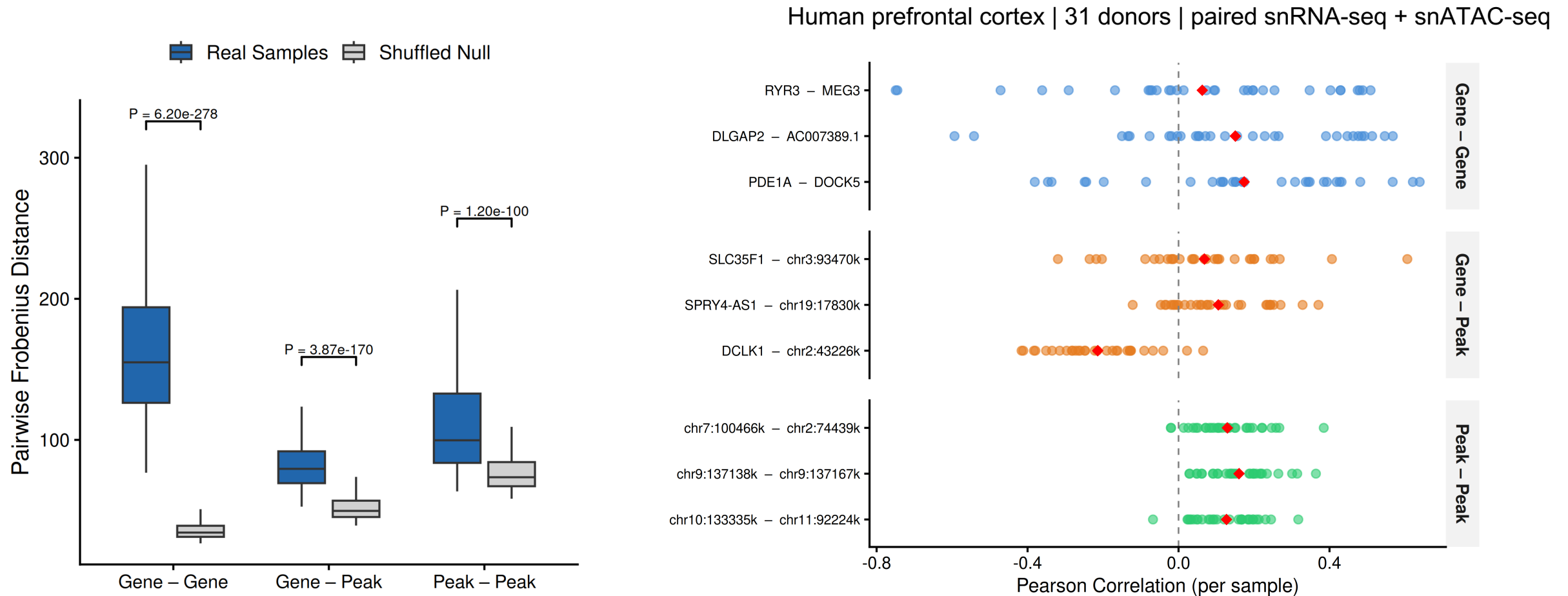
Multi-condition



- Samples stratified by disease, treatment, or timepoint, enabling phenotypic comparison

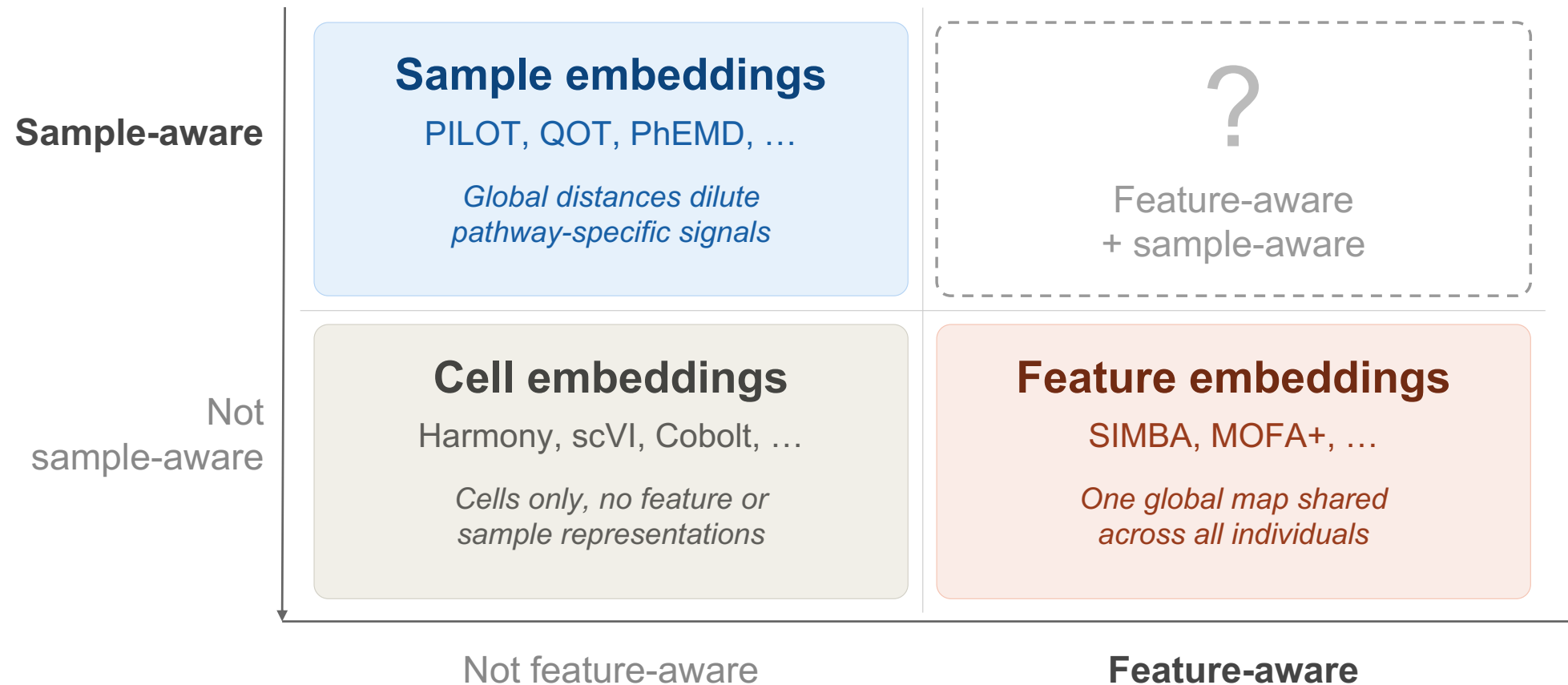
Opportunity: link molecular variation to individual-level phenotypes

Feature relationships vary substantially across individuals



A single global feature representation cannot capture this individual-level variation

Current methods cannot model sample-specific feature relationships



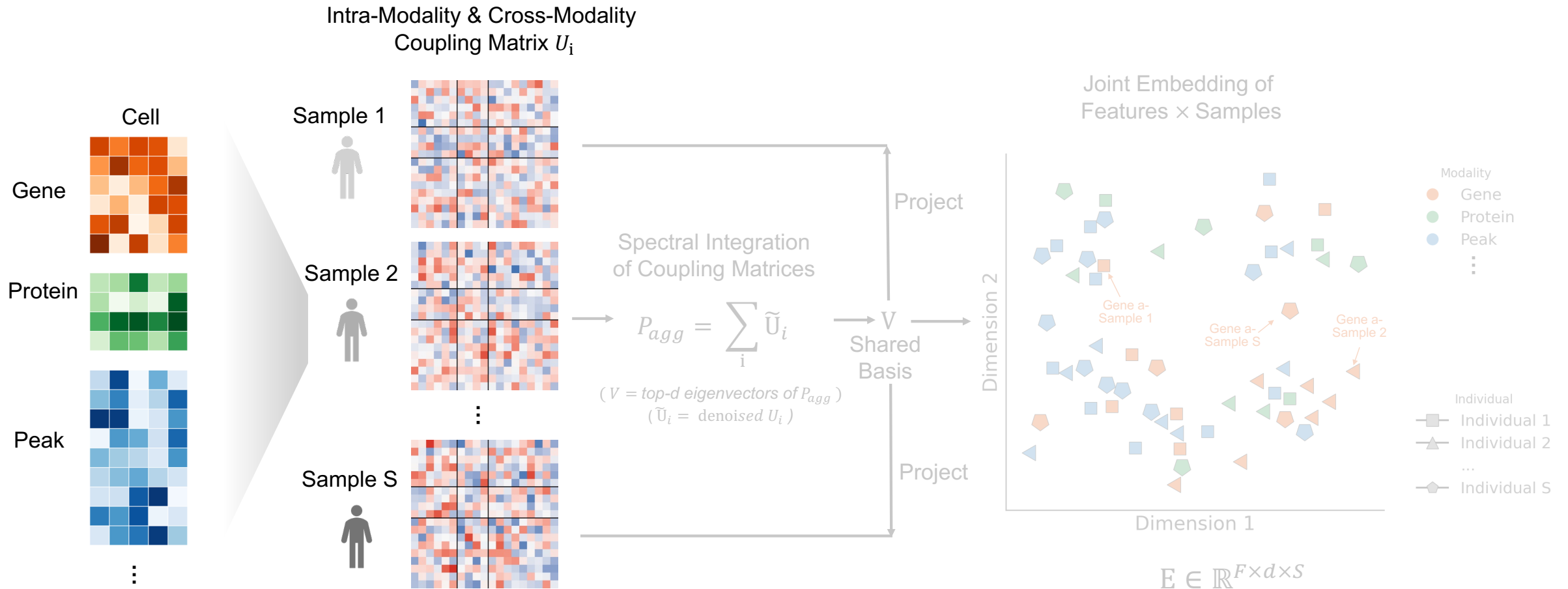
No framework models how feature relationships vary across individuals

MOSAIC: Multi-Omic Sample Wise Analysis of Inter-feature Connectivity

① Per-sample coupling

② Shared spectral basis

③ Feature × sample embedding

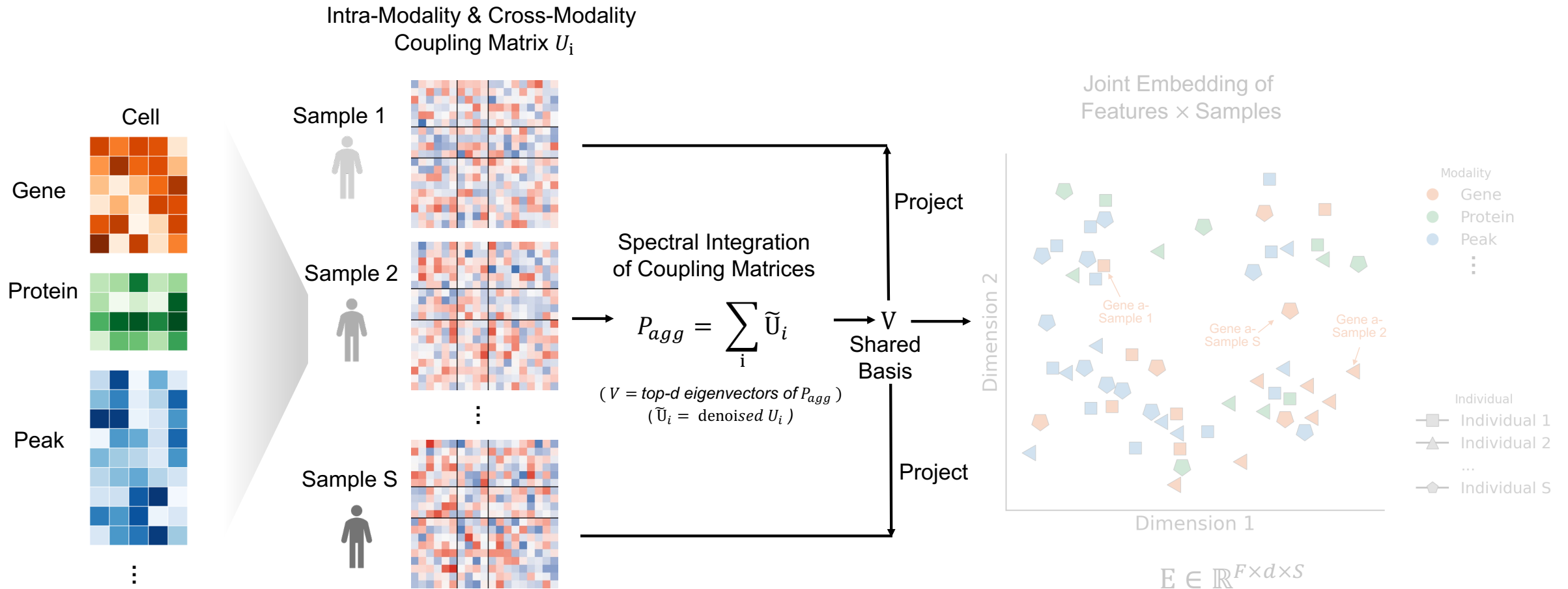


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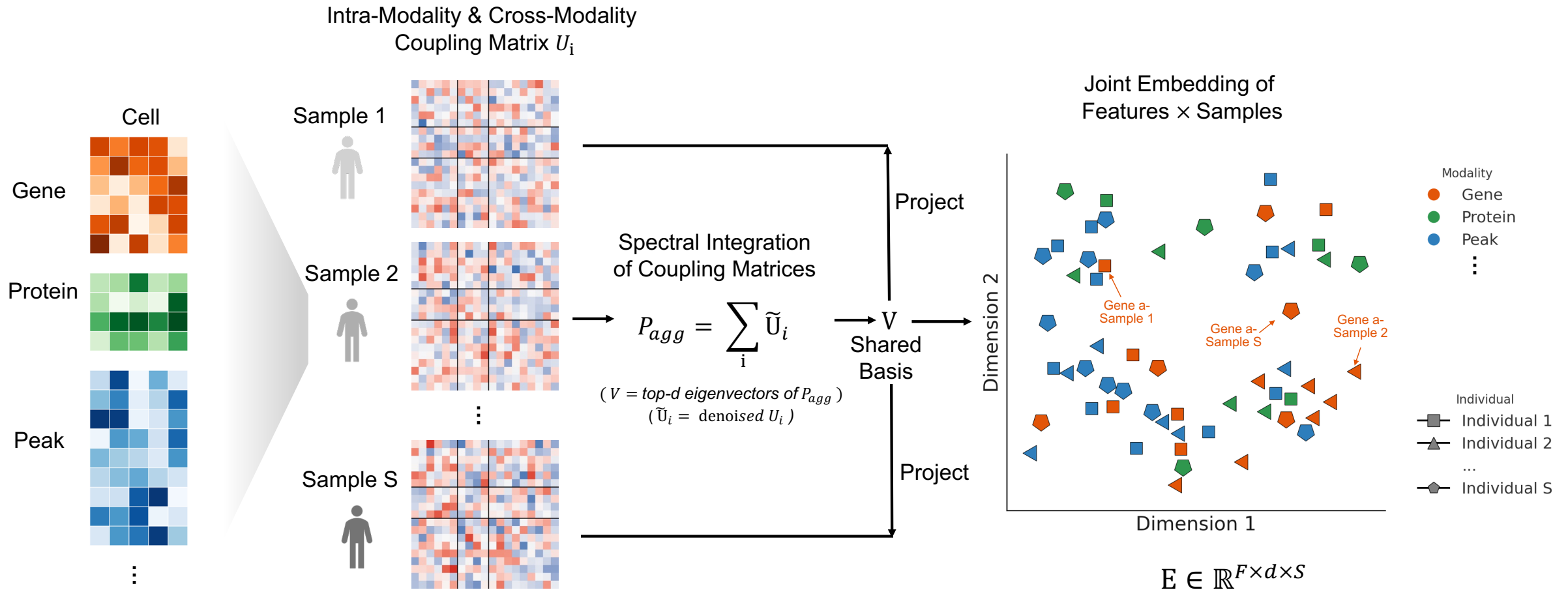


MOSAIC: Multi-Omic Sample Wise Analysis of Inter-feature Connectivity

① Per-sample coupling

② Shared spectral basis

③ Feature × sample embedding



Spectral integration produces natively comparable embeddings

Naive: Embed then align



✗ Each E_i has its own coordinate system; sign/rotation ambiguity makes alignment ill-posed

PCA + Procrustes, MOFA+ + Procrustes, SIMBA + Procrustes

VS

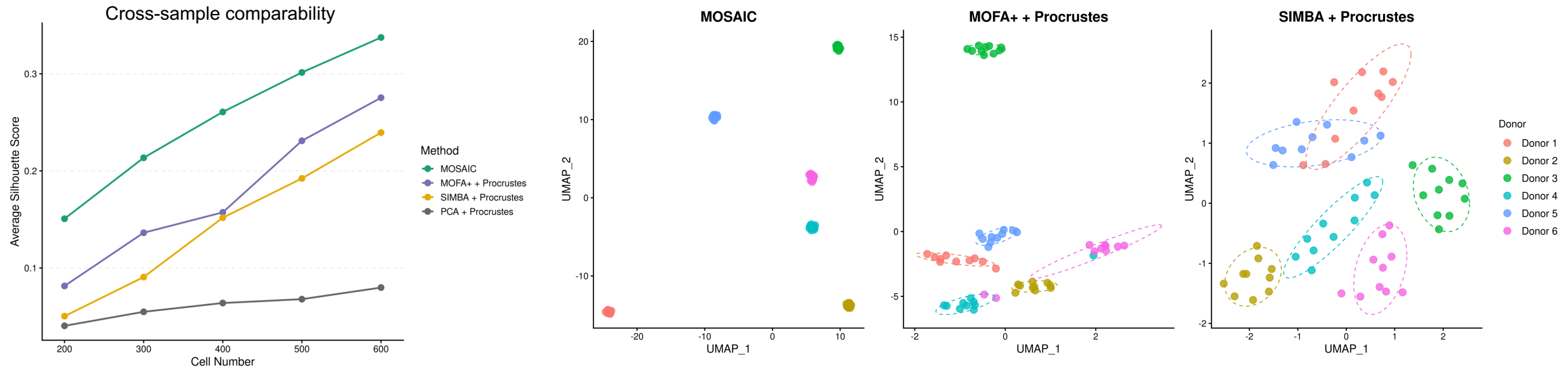
MOSAIC: Joint spectral integration



✓ Shared basis V by construction; structure reinforced, noise averaged out

No post-hoc alignment needed; embeddings are natively comparable

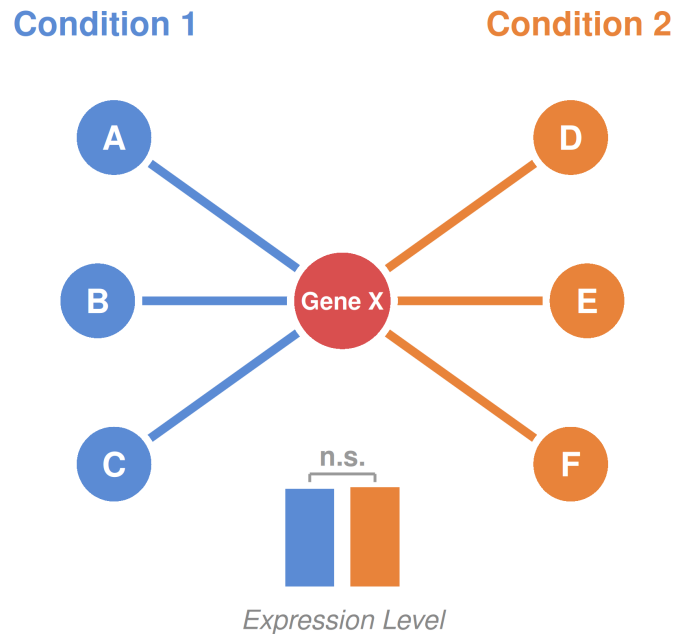
Empirical validation *MOSAIC consistently achieves the highest donor identity recovery*



The joint embedding enables three downstream applications

① Application 1

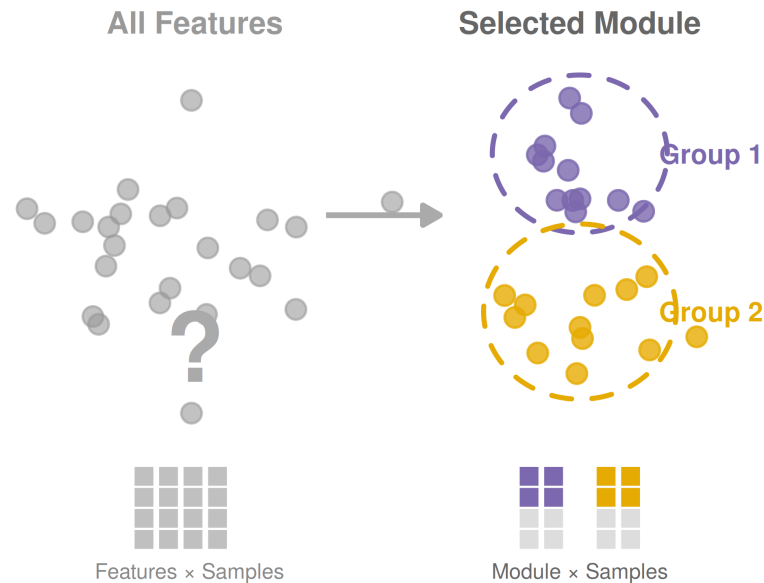
Differential Connectivity



Compare a feature's embedding across conditions

② Application 2

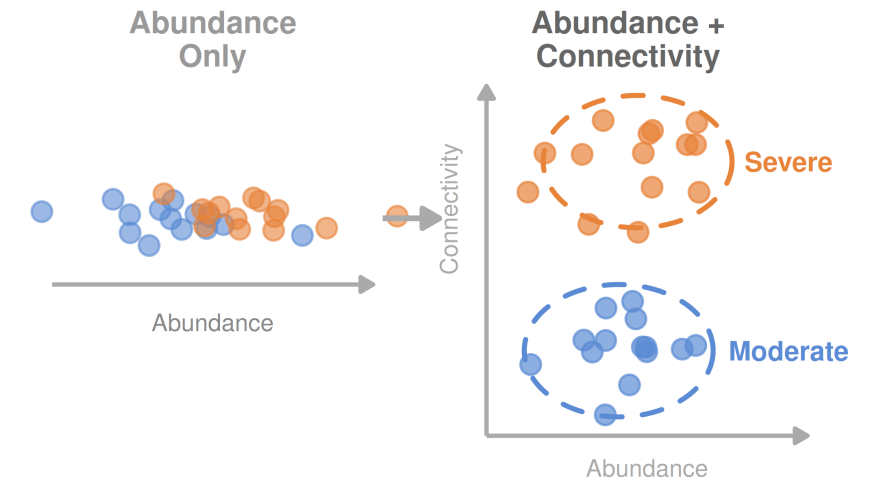
Sample Subgroup Detection



Group features by how they stratify patients

③ Application 3

Clinical Outcome Prediction



Use connectivity as a complementary predictive signal

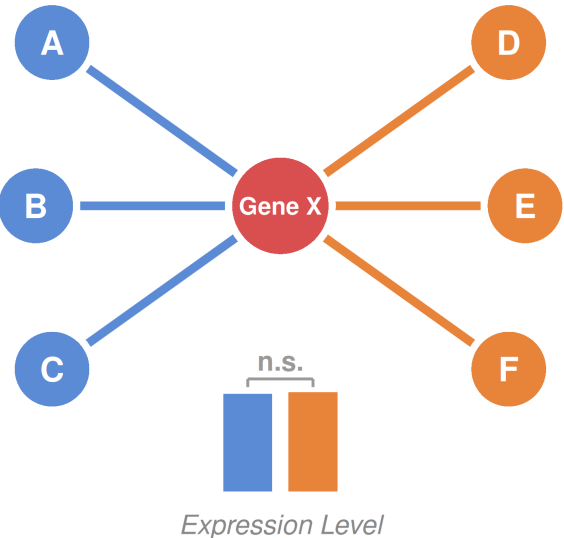
Differential Connectivity: detecting network rewiring invisible to standard methods

① Application 1

Differential Connectivity

Condition 1

Condition 2



Compare a feature's embedding across conditions

② Application 2

Sample Subgroup Detection

All Features

Selected Module

- Standard methods define "differential" by changes in **abundance**
- A feature's function depends on its **regulatory context**, not just its level
- MOSAIC detects features whose network partners change across conditions, even when expression is stable (PERMANOVA on embedding vectors)



Group features by how they stratify patients

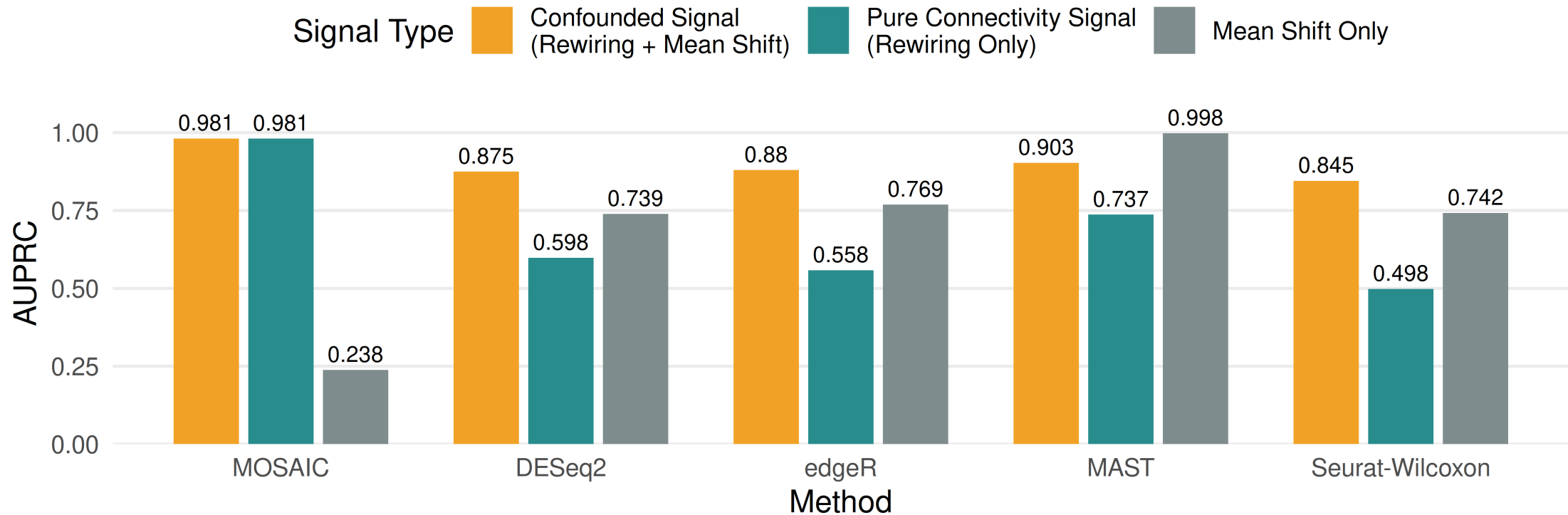
③ Application 3

Clinical Outcome Prediction



Use connectivity as a complementary predictive signal

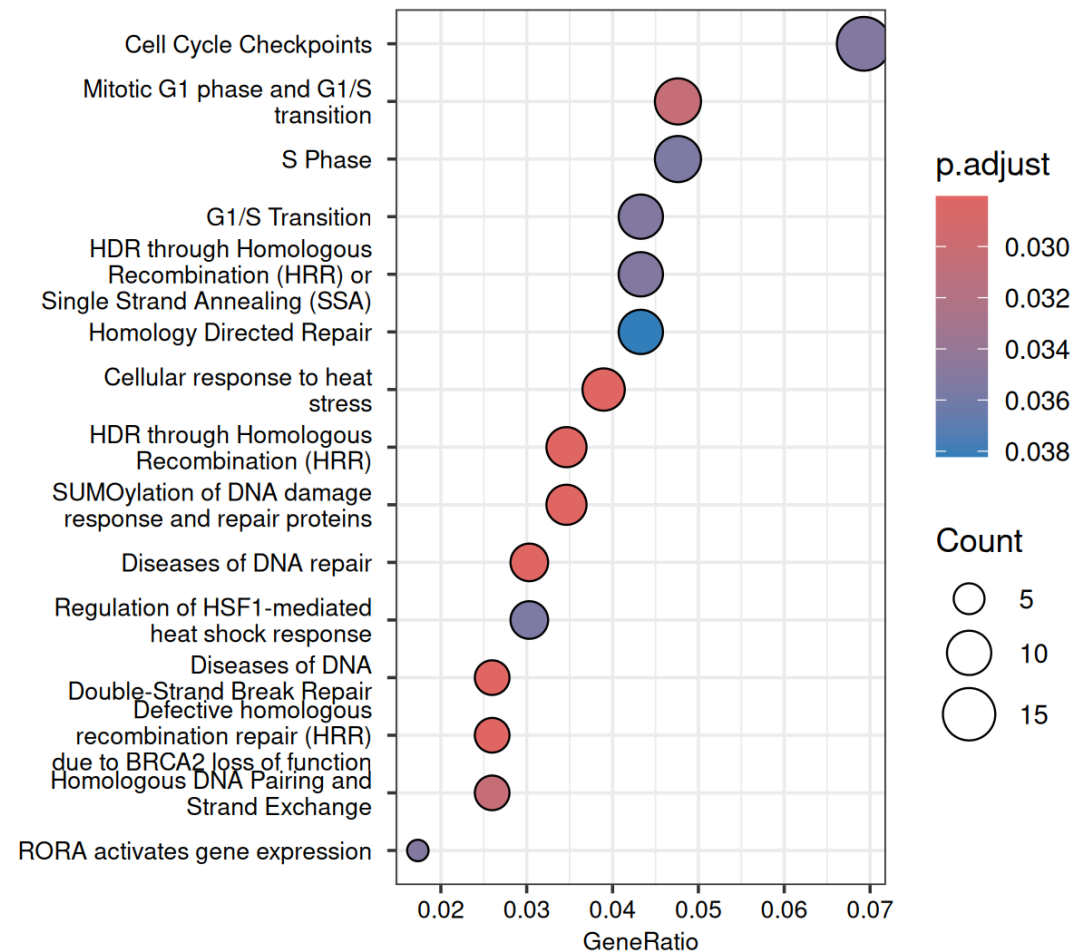
Validating the Signal: Detecting ‘Pure Connectivity’



- Pure Connectivity: MOSAIC achieves AUPRC = 0.981; abundance methods drop to near random chance
- Mean Shift Only: MOSAIC remains near random under mean-shift-only signals, as expected, while abundance methods excel (confirming that connectivity is a complementary, not redundant, signal)
- MOSAIC is not a better abundance detector; it detects a fundamentally different signal

MOSAIC reveals a hidden rewiring program upon T-cell activation

CITE-seq (protein + RNA) | 8 donors | CD4+ Naïve T cells | Day 0 vs Day 7 post-vaccination (Hao et al., Cell, 2021)

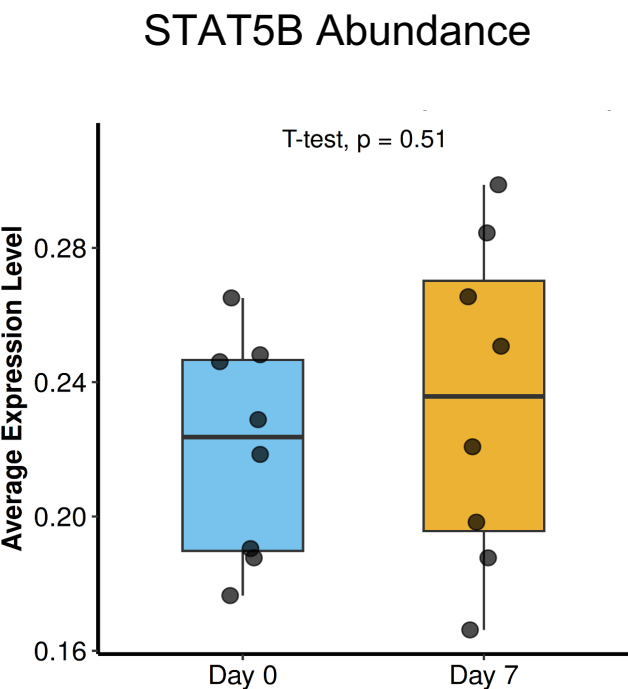


- MOSAIC identified **393 Differential Connectivity (DC) features**, where 392 missed by traditional expression tools (MAST)
- DC features are enriched in cell cycle, DNA repair, and heat stress response pathways
- DC features show significantly higher network turnover between timepoints
(Jaccard $p = 4.3 \times 10^{-15}$)

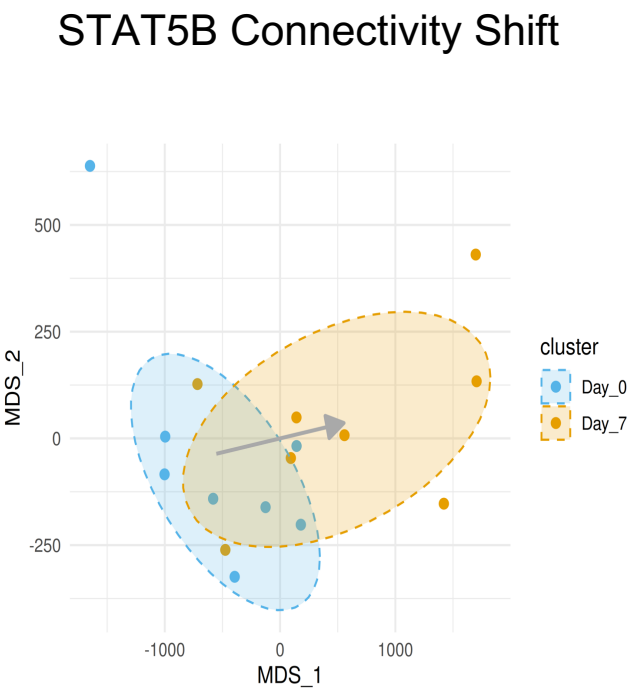
STAT5B rewires its regulatory network without changing expression

DC rank 93, adjusted p = 0.015

Abundance: Unchanged

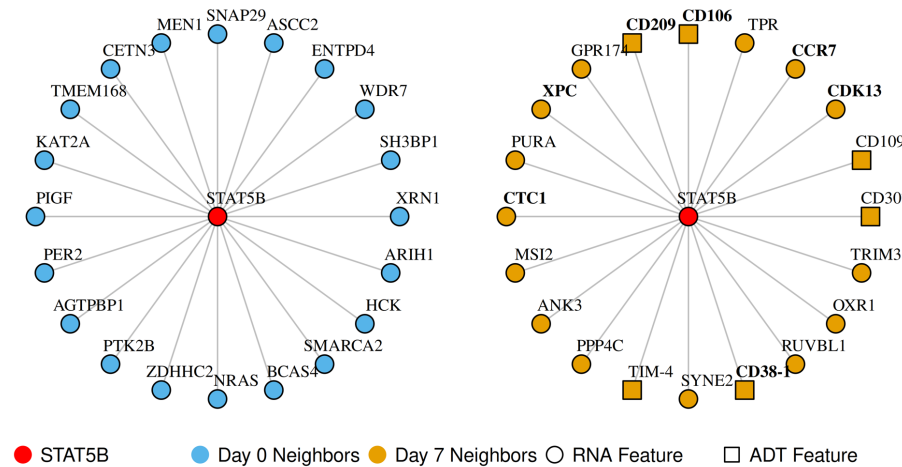


Connectivity: Rewired



The Network

STAT5B Network – Day 0 STAT5B Network – Day 7



Zero overlap in top 20 partners

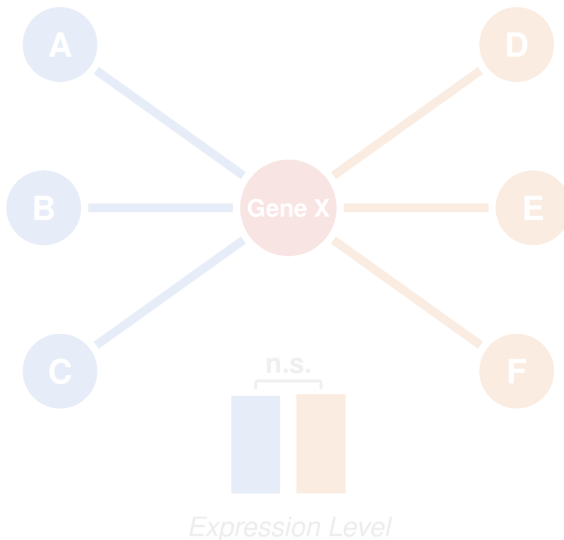
Modular connectivity reveals hidden patient subtypes

① Application 1

Differential Connectivity

Condition 1

Condition 2



Compare a feature's embedding across conditions

② Application 2

Sample Subgroup Detection

All Features

Selected Module



Group features by how they stratify patients

③ Application 3

Clinical Outcome Prediction

- Global patient similarity dilutes disease-relevant signals with noise from thousands of irrelevant features
- MOSAIC groups features into **coherent modules** with coordinated connectivity variation across patients
- Each module is **independently evaluated** for its ability to **partition patients into subgroups**

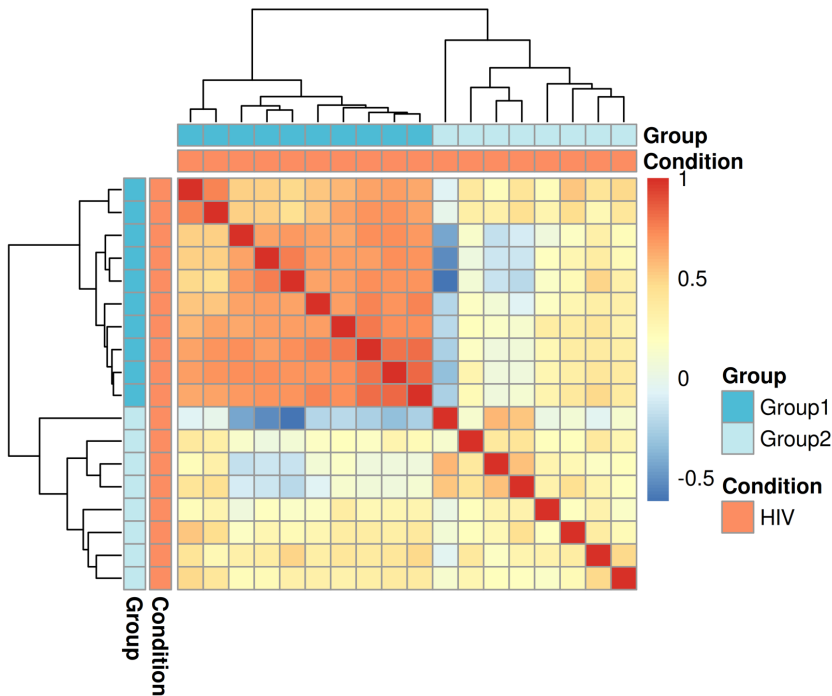
Abundance

Use connectivity as a complementary predictive signal

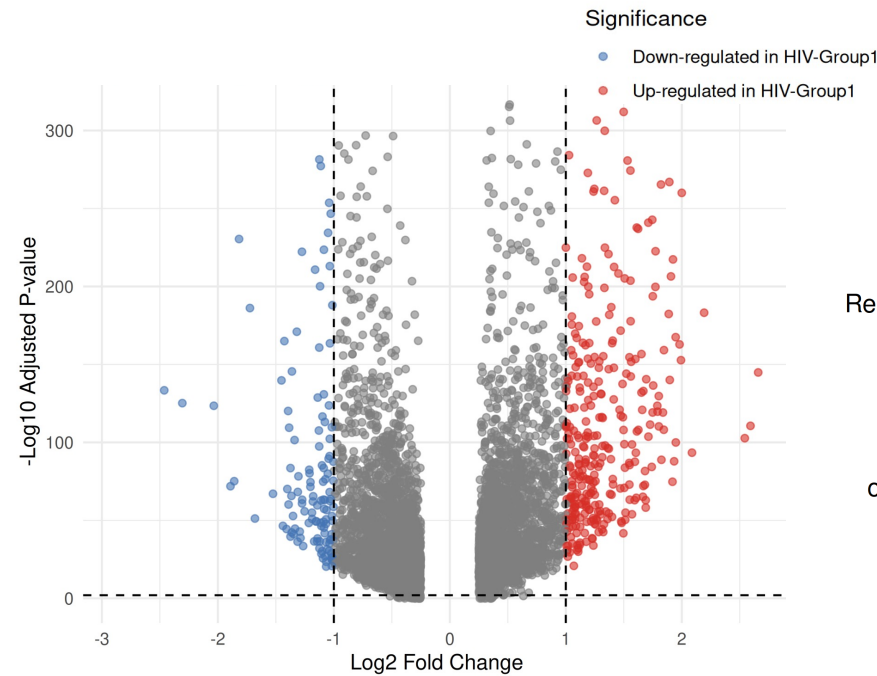
MOSAIC discovers a stress-driven neuronal subtype within HIV+ patients

Prefrontal Cortex | 18 HIV+ samples | snRNA + snATAC | L2/3 Inhibitory Neurons (Yang et al., bioRxiv, 2025)

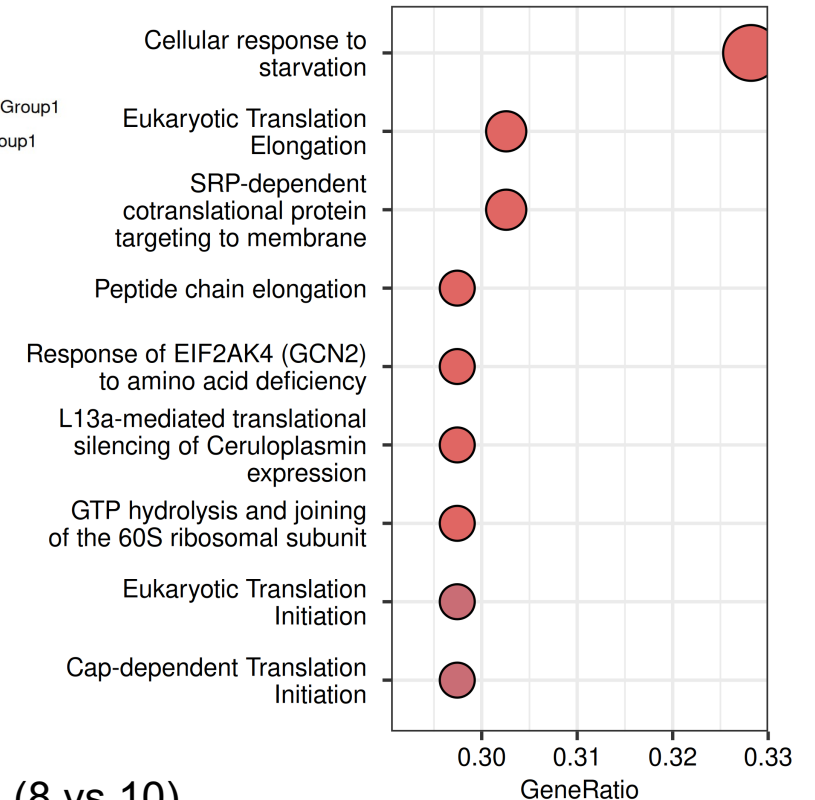
Sample Similarity Heatmap for HIV Subgroups



RNA Differential Expression Between Subgroups



Reactome Pathway Enrichment



- Module 27 (66 features) partitions 18 HIV+ patients into two distinct subgroups (8 vs 10)
- Pathway signature: cellular response to starvation + translation machinery (**Integrated Stress Response**)
- Stress signature replicates in independent brain region (insular cortex, $p < 0.0001$)

Connectivity adds a complementary dimension to clinical prediction

① Application 1

Differential Connectivity

Condition

A

B

C

F

- Standard prediction relies on aggregated expression levels, missing regulatory topology
- MOSAIC connectivity profiles capture a **complementary dimension** of disease variation
- Integrating abundance and connectivity **improves prediction beyond either alone**



Compare a feature's embedding across conditions

② Application 2

Sample Subgroup Detection

Module

Group 1

Group 2

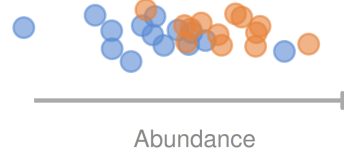


Group features by how they stratify patients

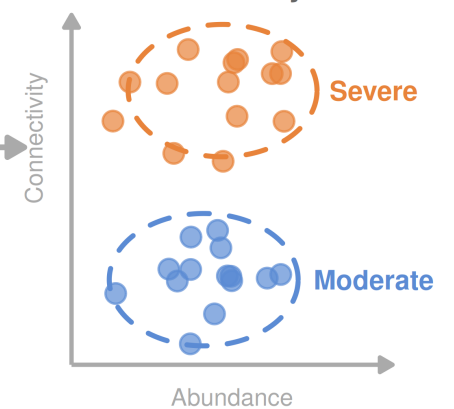
③ Application 3

Clinical Outcome Prediction

Abundance Only



Abundance + Connectivity

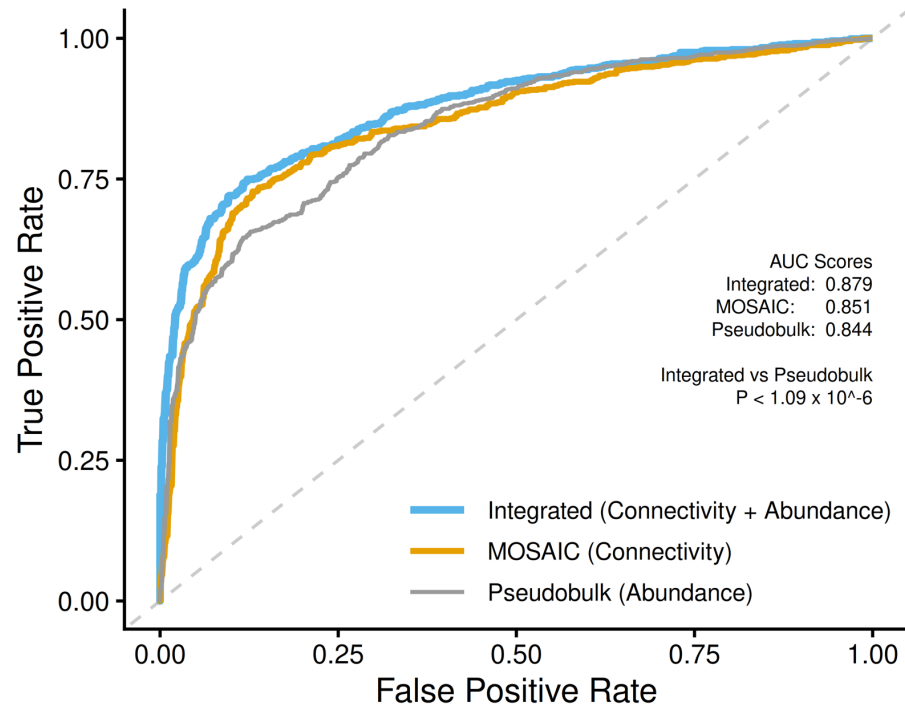


Use connectivity as a complementary predictive signal

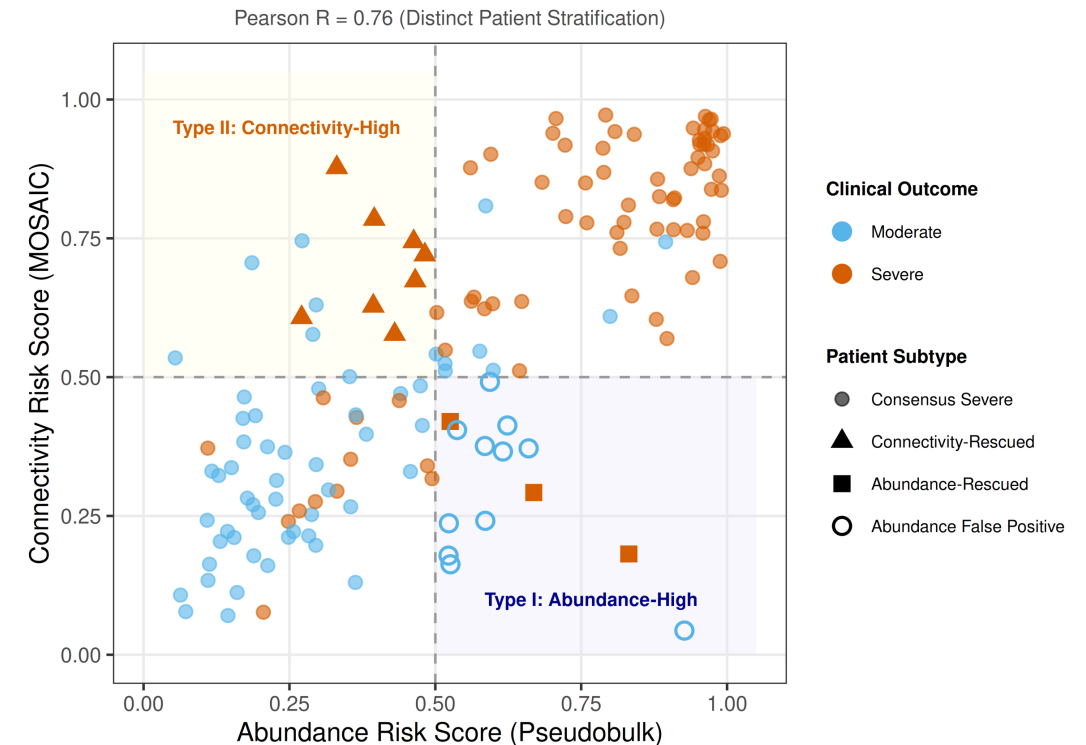
MOSAIC connectivity features improve COVID-19 severity prediction

scRNA-seq | 151 COVID-19 patients | Moderate vs Severe | Monocytes (Ren et al., Cell, 2021)

Predictive Performance for Monocytes



Complementary Mechanisms of Severity

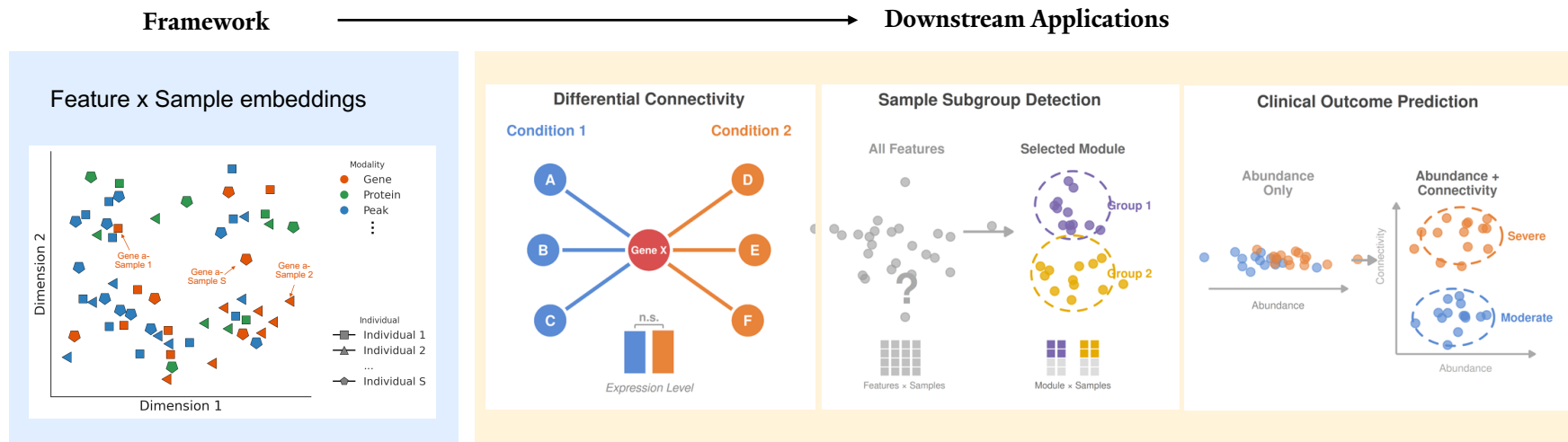


- Integrating connectivity + abundance improves severity prediction (AUC: 0.844 $\rightarrow$ 0.879, $p < 1.09 \times 10^{-6}$)
- Abundance and connectivity rely on largely non-overlapping predictive features (Jaccard = 0.05)
- Connectivity rescues severe patients missed by abundance alone (resolving false negatives)

Summary

Regulatory connectivity is a dimension of variation complementary to, and distinct from, abundance

- MOSAIC makes this dimension accessible by spectral integration of sample-specific coupling matrices into feature $\times$ sample embeddings
- Three applications validate its utility: rewiring detection, patient subtyping, clinical prediction



Acknowledgements



Dr. Rong Ma



Dr. Yuval Kluger



Kluger Lab



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